

VII.

The Theory of Relativity

WITHIN the field of modern physics the theory of relativity has always played a very important role. It was in this theory that the necessity for a change in the fundamental principles of physics was recognized for the first time. Therefore, a discussion of those problems that had been raised and partly solved by the theory of relativity belongs essentially to our treatment of the philosophical implications of modern physics. In some sense it may be said that—contrary to quantum theory—the development of the theory of relativity from the final recognition of the difficulties to their solution has taken only a very short time. The repetition of Michelson's experiment by Morley and Miller in 1904 was the first definite evidence for the impossibility of detecting the translational motion of the earth by optical methods, and Einstein's decisive paper appeared less than two years later. On the other hand, the experiment of Morley and Miller and Einstein's paper were only the final steps in a development which had started very much earlier and which may be summarized under the heading "electrodynamics of moving bodies."

Obviously the electrodynamics of moving bodies had been an important field of physics and engineering ever since electro-motors had been constructed. A serious difficulty had been

brought into this subject, however, by Maxwell's discovery of the electromagnetic nature of light waves. These waves differ in one essential property from other waves, for instance, from sound waves: they can be propagated in what seems to be empty space. When a bell rings in a vessel that has been evacuated, the sound does not penetrate to the outside. But light does penetrate easily through the evacuated volume. Therefore, one assumed that light waves could be considered as elastic waves of a very light substance called ether which could be neither seen nor felt but which filled the evacuated space as well as the space in which other matter, like air or glass, existed. The idea that electromagnetic waves could be a reality in themselves, independent of any bodies, did at that time not occur to the physicists. Since this hypothetical substance ether seemed to penetrate through other matter, the question arose: What happens if the matter is set into motion? Does the ether participate in this motion and—if this is the case—how is a light wave propagated in the moving ether?

Experiments which are relevant to this question are difficult for the following reason: The velocities of moving bodies are usually very small compared to the velocity of light. Therefore, the motion of these bodies can only produce very small effects which are proportional to the ratio of the velocity of the body to the velocity of light, or to a higher power of this ratio. Several experiments by Wilson, Rowland, Roentgen and Eichenwald and Fizeau permitted the measurement of such effects with an accuracy corresponding to the first power of this ratio. The theory of the electrons developed by Lorentz in 1895 was able to describe these effects quite satisfactorily. But then the experiment of Michelson, Morley and Miller created a new situation.

This experiment shall be discussed in some detail. In order to

get bigger effects and thereby more accurate results, it seemed best to do experiments with bodies of very high velocity. The earth moves around the sun with a velocity of roughly 20 miles/sec. If the ether is at rest with respect to the sun and does not move with the earth, then this fast motion of the ether with respect to the earth should make itself felt in a change of the velocity of light. This velocity should be different depending on whether the light is propagated in a direction parallel or perpendicular to the direction of the motion of the ether. Even if the ether should partly move with the earth, there should be some effect due to what one may call wind of the ether, and this effect would then probably depend on the altitude above sea level at which the experiment is carried out. A calculation of the expected effect showed that it should be very small, since it is proportional to the square of the ratio of the velocity of the earth to that of the light, and that one therefore had to carry out very careful experiments on the interference of two beams of light traveling parallel or perpendicular to the motion of the earth. The first experiment of this kind, carried out by Michelson in 1881, had not been sufficiently accurate. But even later repetitions of the experiment did not reveal the slightest signs of the expected effect. Especially the experiments of Morley and Miller in 1904 could be considered as definite proof that an effect of the expected order of magnitude did not exist.

This result, strange as it was, met another point that had been discussed by the physicists some time before. In Newton's mechanics a certain "principle of relativity" is fulfilled that can be described in the following terms: If in a certain system of reference the mechanical motion of bodies fulfills the laws of Newtonian mechanics, then this is also true for any other frame of reference which is in uniform nonrotating motion with respect

to the first system. Or, in other words, a uniform translational motion of a system does not produce any mechanical effects at all and can therefore not be observed by such effects.

Such a principle of relativity—so it seemed to the physicists—could not be true in optics or electrodynamics. If the first system is at rest with respect to the ether, the other systems are not, and therefore their motion with respect to the ether should be recognized by effects of the type considered by Michelson. The negative result of the experiment of Morley and Miller in 1904 revived the idea that such a principle of relativity could be true in electrodynamics as well as Newtonian mechanics.

On the other hand, there was an old experiment by Fizeau in 1851 that seemed definitely to contradict the principle of relativity. Fizeau had measured the velocity of light in a moving liquid. If the principle of relativity was correct, the total velocity of light in the moving liquid should be the sum of the velocity of the liquid and the velocity of light in the liquid at rest. But this was not the case; the experiment of Fizeau showed that the total velocity was somewhat smaller.

Still the negative results of all more recent experiments to recognize the motion “with respect to the ether” inspired the theoretical physicists and mathematicians at that time to look for mathematical interpretations that reconciled the wave equation for the propagation of light with the principle of relativity. Lorentz suggested, in 1904, a mathematical transformation that fulfilled these requirements. He had to introduce the hypothesis that moving bodies are contracted in the direction of motion by a factor depending on the velocity of the body, and in different schemes of reference there are different “apparent” times which in many ways take the place of the “real” time. In this way he could represent something resembling the principle of relativity:

the "apparent" velocity of light was the same in every system of reference. Similar ideas had been discussed by Poincaré, Fitzgerald and other physicists.

The decisive step, however, was taken in the paper by Einstein in 1905 in which he established the "apparent" time of the Lorentz transformation as the "real" time and abolished what had been called "real" time by Lorentz. This was a change in the very foundations of physics; an unexpected and very radical change that required all the courage of a young and revolutionary genius. To take this step one needed, in the mathematical representation of nature, nothing more than the consistent application of the Lorentz transformation. But by its new interpretation the structure of space and time had changed and many problems of physics appeared in a new light. The substance ether, for instance, could be abolished too. Since all systems of reference that are in uniform translation motion with respect to each other are equivalent for the description of nature, there is no meaning in the statement that there is a substance, the ether, which is at rest in only one of these systems. Such a substance is in fact not needed and it is much simpler to say that light waves are propagated through empty space and that electromagnetic fields are a reality of their own and can exist in empty space.

But the decisive change was in the structure of space and time. It is very difficult to describe this change in the words of common language, without the use of mathematics, since the common words "space" and "time" refer to a structure of space and time that is actually an idealization and oversimplification of the real structure. But still we have to try to describe the new structure and we can perhaps do it in the following way:

When we use the term "past" we comprise all those events

which we could know at least in principle, about which we could have heard at least in principle. In a similar manner we comprise by the term "future" all those events which we could influence at least in principle, which we could try to change or to prevent at least in principle. It is not easy for a nonphysicist to see why this definition of the terms "past" and "future" should be the most convenient one. But one can easily see that it corresponds very accurately to our common use of the terms. If we use the terms in this way, it turns out as a result of many experiments that the content of "future" or "past" does not depend on the state of motion or other properties of the observer. We may say that the definition is invariant against the motion of the observer. This is true both in Newtonian mechanics and in Einstein's theory of relativity.

But the difference is this: In classical theory we assume that future and past are separated by an infinitely short time interval which we may call the present moment. In the theory of relativity we have learned that the situation is different: future and past are separated by a finite time interval the length of which depends on the distance from the observer. Any action can only be propagated by a velocity smaller than or equal to the velocity of light. Therefore, an observer can at a given instant neither know of nor influence any event at a distant point which takes place between two characteristic times. The one time is the instant at which a light signal has to be given from the point of the event in order to reach the observer at the instant of observation. The other time is the instant at which a light signal, given by the observer at the instant of the observation, reaches the point of the event. The whole finite time interval between these two instants may be said to belong to the "present time" for the observer at the instant of observation. Any event taking place

between the two characteristic times may be called "simultaneous" with the act of observation.

The use of the phrase "may be called" points up an ambiguity in the word "simultaneous," which is due to the fact that this term has been formed from the experience of daily life, in which the velocity of light can always be considered as infinitely high. Actually this term in physics can be defined also in a slightly different manner and Einstein has in his papers used this second definition. When two events happen at the same point in space simultaneously, we say that they coincide; this term is quite unambiguous. Let us now imagine three points in space that lie on a straight line so that the point in the middle has the same distance from each of the two outer points. If two events happen at the two outer points at such times that light signals starting from the events coincide when they reach the point in the middle, we can define the two events as simultaneous. This definition is narrower than the first one. One of its most important consequences is that when two events are simultaneous for one observer they may not be simultaneous for another observer, if he is in motion relative to the first observer. The connection between the two definitions can be established by the statement that whenever two events are simultaneous in the first sense of the term, one can always find a frame of reference in which they are simultaneous in the second sense too.

The first definition of the term "simultaneous" seems to correspond more nearly to its use in daily life, since the question whether two events are simultaneous does in daily life not depend on the frame of reference. But in both relativistic definitions the term has acquired a precision which is lacking in the language of daily life. In quantum theory the physicists had to learn rather early that the terms of classical physics describe

nature only inaccurately, that their application is limited by the quantum laws and that one therefore should be cautious in their use. In the theory of relativity the physicists have tried to change the meaning of the words of classical physics, to make the terms more precise in such a way that they fit the new situation in nature.

The structure of space and time that has been brought to light by the theory of relativity has many consequences in different parts of physics. The electrodynamics of moving bodies can be derived at once from the principle of relativity. This principle itself can be formulated as a quite general law of nature pertaining not only to electrodynamics or mechanics but to any group of laws: The laws take the same form in all systems of reference, which are different from each other only by a uniform translational motion; they are invariant against the Lorentz transformation.

Perhaps the most important consequence of the principle of relativity is the inertia of energy, or the equivalence of mass and energy. Since the velocity of light is the limiting velocity which can never be reached by any material body, it is easy to see that it is more difficult to accelerate a body that is already moving very fast than a body at rest. The inertia has increased with the kinetic energy. But quite generally any kind of energy will, according to the theory of relativity, contribute to the inertia, i.e., to the mass, and the mass belonging to a given amount of energy is just this energy divided by the square of the velocity of light. Therefore, every energy carries mass with it; but even a rather big energy carries only a very small mass, and this is the reason why the connection between mass and energy had not been observed before. The two laws of the conservation of mass and the conservation of charge lose their separate validity and

are combined into one single law which may be called the law of conservation of energy or mass. Fifty years ago, when the theory of relativity was formulated, this hypothesis of the equivalence of mass and energy seemed to be a complete revolution in physics, and there was still very little experimental evidence for it. In our times we see in many experiments how elementary particles can be created from kinetic energy, and how such particles are annihilated to form radiation; therefore, the transmutation from energy into mass and vice versa suggests nothing unusual. The enormous release of energy in an atomic explosion is another and still more spectacular proof of the correctness of Einstein's equation. But we may add here a critical historical remark.

It has sometimes been stated that the enormous energies of atomic explosions are due to a direct transmutation of mass into energy, and that it is only on the basis of the theory of relativity that one has been able to predict these energies. This is, however, a misunderstanding. The huge amount of energy available in the atomic nucleus was known ever since the experiments of Becquerel, Curie and Rutherford on radioactive decay. Any decaying body like radium produces an amount of heat about a million times greater than the heat released in a chemical process in a similar amount of material. The source of energy in the fission process of uranium is just the same as that in the α -decay of radium, namely, mainly the electrostatic repulsion of the two parts into which the nucleus is separated. Therefore, the energy of an atomic explosion comes directly from this source and is not derived from a transmutation of mass into energy. The number of elementary particles with finite rest mass does not decrease during the explosion. But it is true that the binding energies of the particles in an atomic nucleus do show up in their

masses and therefore the release of energy is in this indirect manner also connected with changes in the masses of the nuclei. The equivalence of mass and energy has—besides its great importance in physics—also raised problems concerning very old philosophical questions. It has been the thesis of several philosophical systems of the past that substance or matter cannot be destroyed. In modern physics, however, many experiments have shown that elementary particles, e.g., positrons and electrons, can be annihilated and transmuted into radiation. Does this mean that these older philosophical systems have been disproved by modern experience and that the arguments brought forward by the earlier systems have been misleading?

This would certainly be a rash and unjustified conclusion, since the terms “substance” and “matter” in ancient or medieval philosophy cannot simply be identified with the term “mass” in modern physics. If one wished to express our modern experience in the language of older philosophies, one could consider mass *and* energy as two different forms of the same “substance” and thereby keep the idea of substance as indestructible.

On the other hand, one can scarcely say that one gains much by expressing modern knowledge in an old language. The philosophic systems of the past were formed from the bulk of knowledge available at their time and from the lines of thought to which such knowledge had led. Certainly one should not expect the philosophers of many hundreds of years ago to have foreseen the development of modern physics or the theory of relativity. Therefore, the concepts to which the philosophers were led in the process of intellectual clarification a long time ago cannot possibly be adapted to phenomena that can only be observed by the elaborate technical tools of our time.

But before going into a discussion of philosophical implica-

tions of the theory of relativity its further development has to be described.

The hypothetical substance "ether," which had played such an important role in the early discussions on Maxwell's theories in the nineteenth century, had—as has been said before—been abolished by the theory of relativity. This is sometimes stated by saying that the idea of absolute space has been abandoned. But such a statement has to be accepted with great caution. It is true that one cannot point to a special frame of reference in which the substance ether is at rest and which could therefore deserve the name "absolute space." But it would be wrong to say that space has now lost all of its physical properties. The equations of motion for material bodies or fields still take a different form in a "normal" system of reference from another one which rotates or is in a nonuniform motion with respect to the "normal" one. The existence of centrifugal forces in a rotating system proves—so far as the theory of relativity of 1905 and 1906 is concerned—the existence of physical properties of space which permit the distinction between a rotating and a nonrotating system.

This may not seem satisfactory from a philosophical point of view, from which one would prefer to attach physical properties only to physical entities like material bodies or fields and not to empty space. But so far as the theory of electromagnetic processes or mechanical motions is concerned, this existence of physical properties of empty space is simply a description of facts that cannot be disputed.

A careful analysis of this situation about ten years later, in 1916, led Einstein to a very important extension of the theory of relativity, which is usually called the theory of "general relativity." Before going into a description of the main ideas of this new theory it may be useful to say a few words about the degree

of certainty with which we can rely on the correctness of these two parts of the theory of relativity. The theory of 1905 and 1906 is based on a very great number of well-established facts: on the experiments of Michelson and Morley and many similar ones, on the equivalence of mass and energy in innumerable radioactive processes, on the dependence of the lifetime of radioactive bodies on their velocity, etc. Therefore, this theory belongs to the firm foundations of modern physics and cannot be disputed in our present situation.

For the theory of general relativity the experimental evidence is much less convincing, since the experimental material is very scarce. There are only a few astronomical observations which allow a checking of the correctness of the assumptions. Therefore, this whole theory is more hypothetical than the first one.

The cornerstone of the theory of general relativity is the connection between inertia and gravity. Very careful measurements have shown that the mass of a body as a source of gravity is exactly proportional to the mass as a measure for the inertia of the body. Even the most accurate measurements have never shown any deviation from this law. If the law is generally true, the gravitational forces can be put on the same level with the centrifugal forces or with other forces that arise as a reaction of the inertia. Since the centrifugal forces had to be considered as due to physical properties of empty space, as had been discussed before, Einstein turned to the hypothesis that the gravitational forces also are due to properties of empty space. This was a very important step which necessitated at once a second step of equal importance. We know that the forces of gravity are produced by masses. If therefore gravitation is connected with properties of space, these properties of space must be caused or influenced by the masses. The centrifugal forces in a rotating system must be

produced by the rotation (relative to the system) of possibly very distant masses.

In order to carry out the program outlined in these few sentences Einstein had to connect the underlying physical ideas with the mathematical scheme of general geometry that had been developed by Riemann. Since the properties of space seemed to change continuously with the gravitational fields, its geometry had to be compared with the geometry on curved surfaces where the straight line of Euclidean geometry has to be replaced by the geodetical line, the line of shortest distance, and where the curvature changes continuously. As a final result Einstein was able to give a mathematical formulation for the connection between the distribution of masses and the determining parameters of the geometry. This theory did represent the common facts about gravitation. It was in a very high approximation identical with the conventional theory of gravitation and predicted furthermore a few interesting effects which were just at the limit of measurability. There was, for instance, the action of gravity on light. When monochromatic light is emitted from a heavy star, the light quanta lose energy when moving away through the gravitational field of the star; a red shift of the emitted spectral line follows. There is as yet no experimental evidence for this red shift, as the discussion of the experiments by Freundlich has clearly shown. But it would also be premature to conclude that the experiments contradict the prediction of Einstein's theory. A beam of light that passes near the sun should be deflected by its gravitational field. The deflection has been found experimentally by Freundlich in the right order of magnitude; but whether the deflection agrees quantitatively with the value predicted by Einstein's theory has not yet been decided. The best evidence for the validity of the theory of general relativity seems

to be the procession in the orbital motion of the planet Mercury, which apparently is in very good agreement with the value predicted by the theory.

Though the experimental basis of general relativity is still rather narrow, the theory contains ideas of the greatest importance. During the whole period from the mathematicians of ancient Greece to the nineteenth century, Euclidean geometry had been considered as evident; the axioms of Euclid were regarded as the foundation of any mathematical geometry, a foundation that could not be disputed. Then, in the nineteenth century, the mathematicians Bolyai and Lobachevsky, Gauss and Riemann found that other geometries could be invented which could be developed with the same mathematical precision as that of Euclid; therefore, the questions as to which geometry was correct turned out to be an empirical one. But it was only through the work of Einstein that the question could really be taken up by the physicists. The geometry discussed in the theory of general relativity was not concerned with three-dimensional space only but with the four-dimensional manifold consisting of space and time. The theory established a connection between the geometry in this manifold and the distribution of masses in the world. Therefore, this theory raised in an entirely new form the old questions of the behavior of space and time in the largest dimensions; it could suggest possible answers that could be checked by observations.

Consequently, very old philosophic problems were taken up that had occupied the mind of man since the earliest phases of philosophy and science. Is space finite or infinite? What was there before the beginning of time? What will happen at the end of time? Or is there no beginning and no end? These questions had found different answers in different philosophies and re-

ligions. In the philosophy of Aristotle, for instance, the total space of the universe was finite (though it was infinitely divisible). Space was due to the extension of bodies, it was connected with the bodies; there was no space where there were no bodies. The universe consisted of the earth and the sun and the stars: a finite number of bodies. Beyond the sphere of the stars there was no space; therefore, the space of the universe was finite.

In the philosophy of Kant this question belonged to what he called "antinomies"—questions that cannot be answered, since two different arguments lead to opposite results. Space cannot be finite, since we cannot imagine that there should be an end to space; to whichever point in space we come we can always imagine that we can go beyond. At the same time space cannot be infinite, because space is something that we can imagine (else the word "space" would not have been formed) and we cannot imagine an infinite space. For this second thesis the argument of Kant has not been verbally reproduced. The sentence "space is infinite" means for us something negative; we cannot come to an end of space. For Kant it means that the infinity of space is really given, that it "exists" in a sense that we can scarcely reproduce. Kant's result is that a rational answer to the question whether space is finite or infinite cannot be given because the whole universe cannot be the object of our experience.

A similar situation is found with respect to the problem of the infinity of time. In the *Confessions* of St. Augustine, for instance, this question takes the form: What was God doing before He created the world? Augustine is not satisfied with the joke: "God was busy preparing Hell for those who ask foolish questions." This, he says, would be too cheap an answer, and he tries to give a rational analysis of the problem. Only for us is time passing by; it is expected by us as future; it passes by as the

present moment and is remembered by us as past. But God is not in time; a thousand years are for Him as one day, and one day as a thousand years. Time has been created together with the world, it belongs to the world, therefore time did not exist before the universe existed. For God the whole course of the universe is given at once. There was no time before He created the world. It is obvious that in such statements the word "created" at once raises all the essential difficulties. This word as it is usually understood means that something has come into being that has not been before, and in this sense it presupposes the concept of time. Therefore, it is impossible to define in rational terms what could be meant by the phrase "time has been created." This fact reminds us again of the often discussed lesson that has been learned from modern physics: that every word or concept, clear as it may seem to be, has only a limited range of applicability.

In the theory of general relativity these questions about the infinity of space and time can be asked and partly answered on an empirical basis. If the connection between the four-dimensional geometry in space and time and the distribution of masses in the universe has been correctly given by the theory, then the astronomical observations on the distribution of galaxies in space give us information about the geometry of the universe as a whole. At least one can build "models" of the universe, cosmological pictures, the consequences of which can be compared with the empirical facts.

From the present astronomical knowledge one cannot definitely distinguish between several possible models. It may be that the space filled by the universe is finite. This would not mean that there is an end of the universe at some place. It would only mean that by proceeding farther and farther in one direction in the universe one would finally come back to the point

from which one had started. The situation would be similar as in the two-dimensional geometry on the surface of the earth where we, when starting from a point in an eastward direction, finally come back to this point from the west.

With respect to time there seems to be something like a beginning. Many observations point to an origin of the universe about four billion years ago; at least they seem to show that at that time all matter of the universe was concentrated in a much smaller space than it is now and has expanded ever since from this small space with different velocities. The same time of four billion years is found in many different observations (e.g., from the age of meteorites, of minerals on the earth, etc.), and therefore it would be difficult to find an interpretation essentially different from this idea of an origin. If it is the correct one it would mean that beyond this time the concept of time would undergo essential changes. In the present state of astronomical observations the questions about the space-time geometry on a large scale cannot yet be answered with any degree of certainty. But it is extremely interesting to see that these questions may possibly be answered eventually on a solid empirical basis. For the time being even the theory of general relativity rests on a very narrow experimental foundation and must be considered as much less certain than the so-called theory of special relativity expressed by the Lorentz transformation.

Even if one limits the further discussions of this latter theory there is no doubt that the theory of relativity has deeply changed our views on the structure of space and time. The most exciting aspect of these changes is perhaps not their special nature but the fact that they have been possible. The structure of space and time which had been defined by Newton as the basis of his mathematical description of nature was simple and consistent and corresponded very closely to the use of the concepts space

and time in daily life. This correspondence was in fact so close that Newton's definitions could be considered as the precise mathematical formulation of these common concepts. Before the theory of relativity it seemed completely obvious that events could be ordered in time independent of their location in space. We know now that this impression is created in daily life by the fact that the velocity of light is so very much higher than any other velocity occurring in practical experience; but this restriction was of course not realized at that time. And even if we know the restriction now we can scarcely imagine that the time order of events should depend on their location.

The philosophy of Kant later on drew attention to the fact that the concepts of space and time belong to our relation to nature, not to nature itself; that we could not describe nature without using these concepts. Consequently, these concepts are "a priori" in some sense, they are the condition for and not primarily the result of experience, and it was generally believed that they could not be touched by new experience. Therefore, the necessity of the change appeared as a great surprise. It was the first time that the scientists learned how cautious they had to be in applying the concepts of daily life to the refined experience of modern experimental science. Even the precise and consistent formulation of these concepts in the mathematical language of Newton's mechanics or their careful analysis in the philosophy of Kant had offered no protection against the critical analysis possible through extremely accurate measurements. This warning later proved extremely useful in the development of modern physics, and it would certainly have been still more difficult to understand quantum theory had not the success of the theory of relativity warned the physicists against the uncritical use of concepts taken from daily life or from classical physics.